MATHEMATICAL OPERATIONS

Balancing Equations with Signs and Interchanged Symbols

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 24

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: a sign-replacement (balance the equation) question appears in English 2022 (Q57).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). One mathematical-operations question appears (English 2022 Q57). It is solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The one PYQ and what it asks
2	Foundation	BODMAS, sign-replacement method, interchanged signs
3	Solved PYQs	English 2022 Q57 with step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

One question, in English 2022

Paper	Q.	Question	Type	Ans
English 2022	57	Select the correct combination of signs that can sequentially replace the * signs and balance the equation: 72 * 12 * 5 * 35 * 7 * 58	Balance with signs and =	A
English 2019 / Mathematics 2024 / Science 2024 / Social Studies 2023	–	none		

PYQ INSIGHT: The PYQ gives six numbers and four options, each a list of five signs that includes one **= sign**. Only the correct option makes the left side equal to the right side after BODMAS. Because each option is a full list, you can **test the options one by one**.

Pattern	PYQs	Priority
Balance the equation by choosing the signs (including =)	1	★★★ Must
Interchanged signs (“if + means ×”)	0	★★ Expected
Find the value after replacing symbols	0	★★ Expected

CHAPTER 2

Foundation

BODMAS and a fixed method

2.1 BODMAS

Letter	Operation	Order
B	Brackets	1st
O	Of (also powers/orders)	2nd
D / M	Division and multiplication	3rd (left to right)
A / S	Addition and subtraction	4th (left to right)

NOTE: Division and multiplication have equal rank and are done left to right; so do addition and subtraction.

2.2 Balancing the equation (the PYQ type)

Step	Do this
1	Note where the = sign sits in each option; the two sides must be equal.
2	Substitute the signs of the first option and evaluate each side by BODMAS.
3	If the sides are unequal, go to the next option; stop at the first that balances.
4	Use number sense to shortcut: for example, a side that is much larger than the other cannot be fixed by + and −.

2.3 Interchanged signs

Step	Do this
1	Write the real operation under each displayed symbol.
2	Rewrite the expression with the real operations.
3	Evaluate using BODMAS (not left to right!).

SHORTCUT: English 2022 Q57: 72 ÷ 12 × 5 + 35 − 7 = 58 gives 6 × 5 = 30; 30 + 35 = 65; 65 − 7 = 58. The left side is 58 and the right side is 58.

CHAPTER 3

Solved PYQs

English 2022 Q57 with step-by-step reasoning

English 2022 Q57

Select the correct combination of mathematical signs that can sequentially replace the * signs and balance the given equation: 72 * 12 * 5 * 35 * 7 * 58. (A) ÷, ×, +, −, = (B) +, +, −, =, × (C) =, ×, +, +, − (D) ×, +, =, −, +

1. Try option A: 72 ÷ 12 × 5 + 35 − 7 = 58.
2. Left side: 72 ÷ 12 = 6; 6 × 5 = 30; 30 + 35 = 65; 65 − 7 = 58. Right side: 58. Both sides are equal.
3. Option B: 72 + 12 + 5 − 35 = 7 × 58 gives 54 on the left and 406 on the right (not equal).
4. Option C: 72 = 12 × 5 + 35 + 7 − 58 gives 72 on the left and 4 on the right. Option D: 72 × 12 + 5 = 35 − 7 + 58 gives 869 on the left and 86 on the right. Only A balances.

Answer: Option A

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Test the options** rather than building the equation yourself.
- **Size check:** × and ÷ change values the most; the side with the large number needs the larger result.
- **BODMAS first, then left to right.**
- **Interchanged signs:** replace every symbol, do not rely on memory of the original meaning.
- **Time:** 45 seconds per question.

Trap	How to avoid it
Evaluating left to right ignoring BODMAS	5 + 3 × 2 = 11, not 16.
Forgetting the = position	It splits the equation into two sides.
Using the original meaning of a swapped sign	Replace every symbol first.
Division giving a fraction	Keep fractions; the sides must be exactly equal.

Scope

Related PYQs / topics	Where
Series and coding questions with numbers	Series / Coding (earlier topics)
Number-digit puzzles	Alphanumeric Series and Number Puzzles (Topic 25)

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. Balance the equation

Q1. Select the correct combination of signs that can sequentially replace the * signs and balance the equation: 2 * 7 * 9 * 11 * 12

A. ×, ÷, =, ÷ B. ×, +, =, + C. +, −, =, ÷ D. −, ×, =, +

Q2. Select the correct combination of signs that can sequentially replace the * signs and balance the equation: 9 * 8 * 4 * 10 * 5

A. +, ×, =, − B. ×, −, +, = C. ÷, ÷, ÷, = D. −, +, =, −

Q3. Select the correct combination of signs that can sequentially replace the * signs and balance the equation: 7 * 6 * 9 * 11 * 3

A. ×, =, +, × B. −, =, ×, − C. +, =, ×, − D. +, =, ×, +

B. Interchanged signs

Q4. If + means ×, − means ÷, × means +, ÷ means −, what is the value of 12 + 3 − 4 × 5 ÷ 2?

A. 5 B. 14 C. 12 D. 9

Q5. If + means −, − means +, × means ÷, ÷ means ×, what is the value of 20 × 4 + 3 ÷ 6 − 2?

A. -11 B. -9 C. -14 D. -4

Q6. If + means ×, − means +, × means −, ÷ means ÷, what is the value of 6 + 4 × 10 − 6 ÷ 3?

A. 44 B. 18 C. 13 D. 16

Answer key

Q	Answer	Reason
1	B	Correct equation: 2 × 7 + 9 = 11 + 12 (checked by program); each other combination fails to balance.
2	D	Correct equation: 9 − 8 + 4 = 10 − 5 (checked by program); each other combination fails to balance.
3	A	Correct equation: 7 × 6 = 9 + 11 × 3 (checked by program); each other combination fails to balance.
4	C	Replace each symbol by its real operation: 12 × 3 ÷ 4 + 5 − 2 = 12 (BODMAS).
5	A	Replace each symbol by its real operation: 20 ÷ 4 − 3 × 6 + 2 = -11 (BODMAS).
6	D	Replace each symbol by its real operation: 6 × 4 − 10 + 6 ÷ 3 = 16 (BODMAS).

CHAPTER 6

Quick Revision

One page for the last day

Item	Rule
BODMAS	Brackets, Of, Division and Multiplication, Addition and Subtraction
Balance	Evaluate both sides of the = sign
Interchanged signs	Replace the symbols, then use BODMAS
Option testing	Check the options one by one

PYQ RULE: English 2022 Q57: only option A balances: 72 ÷ 12 × 5 + 35 − 7 = 58.

ROUTINE: Exam-day routine: place the = sign; apply BODMAS to each side; test options; for swapped signs, rewrite first. 45 seconds per question.